Edge-First Image Analysis for Digital Pathology

Technical report, benchmark study and interview guide for two projects:
VAE-Triage (gigapixel WSI triage) and *Satellite_based_tile_segmentation* (high-resolution segmentation)

Prepared for the AIRA Matrix interview

All numbers in this report were measured in this study unless marked "literature".

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Chapter 1

Executive summary (read this first)

1.1 What the two projects are, in one paragraph each

VAE-Triage (pathology). A whole-slide image (WSI) of an H&E-stained breast lymph-node section is a ∼100,000 × 200,000-pixel pyramidal TIFF. Most of it is empty glass, and some tissue tiles are unusable (out of focus, pen ink, folds, bubbles). Running a heavy segmentation network on every tile wastes GPU time and produces garbage on artefacts. VAE-Triage is a *gate in front of the heavy model*: it tiles the slide lazily, throws away glass, flags bad tiles and keeps a compact description of the good tiles. In this study we rebuilt it as an **edge-first, Gaussian-Mixture-Model (GMM) pipeline** that runs on a CPU next to the scanner, and we kept the original VQ-VAE as an optional deep component.

Satellite_based_tile_segmentation (segmentation). Semantic segmentation of 6000×6000-pixel aerial orthophotos (ISPRS Potsdam, 6 classes) with a residual U-Net with a dilated (ASPP-like) bottleneck, patch-based training, overlap stitching and 8× test-time augmentation. The *engineering problem is identical to WSI segmentation*: an image too large for the network is cut into patches, predicted, and stitched back. This is the project to talk about when the interviewer asks "how do you do segmentation?".

1.2 Pipeline at a glance

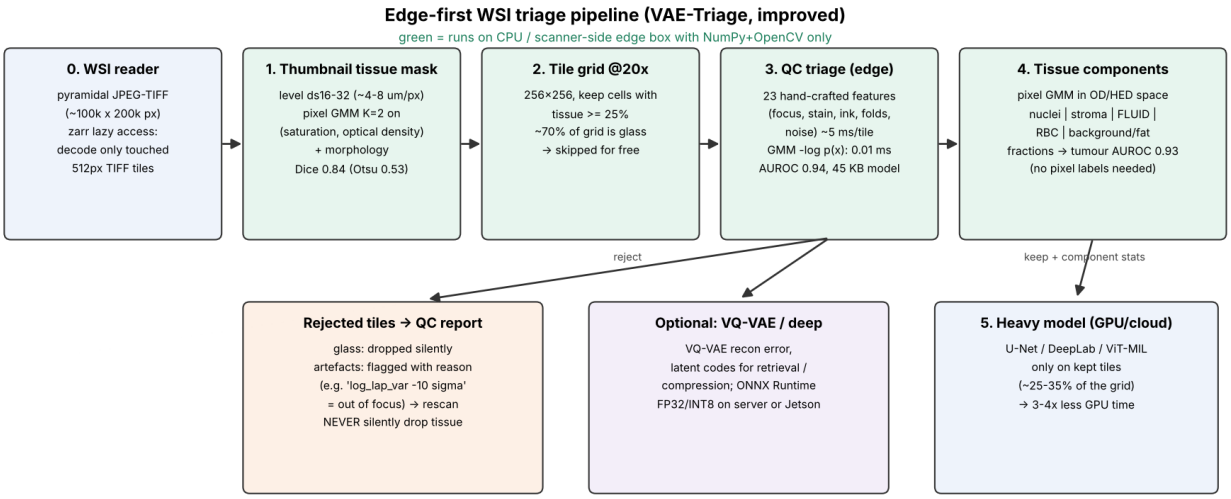


Figure 1.1: The improved VAE-Triage pipeline. Everything in green runs on a CPU with NumPy + OpenCV only.

1.3 Headline numbers (all measured in this study)

Data: 7 real CAMELYON16 breast lymph-node WSIs (4.5 GB, 203,168 tiles at 20×), expert tumour masks; ISPRS Potsdam RGB+NIR, 6 training tiles + 3 official test tiles. CPU only (4 vCPU), no GPU.

Task	**Best classical / ours**	**Reference points**
Stage 0: tissue vs glass (Dice)	**GMM K=2 on (S, OD): 0.839**	Otsu 0.52–0.55; fixed thr. 0.74; k-means 0.14
Compute removed before any NN	**69.6% of 203k tiles are glass**	30–78% per slide
Stage 1: tile QC (AUROC / F1)	**GMM 0.939 / 0.897**, single Gaussian 0.954 / 0.906	Laplacian 0.76; HistoQC-style rules 0.86; IsolationForest 0.91; supervised LR/RF 0.96–0.97
Tumour wrongly rejected by QC	**3.5%** with slide-level calibration	43% with naive same-slide calibration (a trap!)
Original VQ-VAE MSE as QC score	AUROC **0.376** (worse than chance)	blur/glass reconstruct *better* than tissue
Supervised MobileNetV3 QC	AUROC 0.976, 0.6 ms/tile (ORT)	but recall 7–64% on artefact types not in training; GMM 64–99%
QC model size / cost	**45 KB JSON, 1,799 numbers, 0.01 ms/tile** + 2.1 ms features (128 px)	VQ-VAE 0.69 M params, 22.9 ms/tile ORT
Tumour vs normal, no tumour labels	GMM novelty 0.740; VQ-VAE 0.749	Lit. plain AE 0.66, L1-AE 0.21, perceptual AE 0.93
Tumour via GMM tissue components (fluid, nuclei, stroma)	**0.931** (5 numbers per tile + LR)	supervised RF on 23 features 0.945
Potsdam segmentation, classical (mIoU, 5 cls)	GMM-Bayes + smoothing **45.9**; RF+NIR 48.4	k-means 15.7; lit. deep SOTA 86.5–87.5

Table 1.1: Summary. Deep-learning rows and the time-boxed U-Net / YOLO runs are in the chapters and in the auto-generated tables.

1.4 The six things the candidate must be able to say

1. **Why triage at all:** 70% of a slide is glass and a few % is unusable; the GPU model should only see the remaining ∼30%. That is a 3–4× saving before any clever model.
2. **Why a GMM and not a deep net for this gate:** the decision "is this tile normal-looking" is a *density estimation* problem; a GMM on 23 physically meaningful features gives 0.94 AUROC with a 45 KB model, microsecond scoring, no GPU, no labels, and a human-readable reason for every rejection.
3. **Why a GMM and not k-means or Otsu:** k-means is a GMM with equal spherical covariances and hard assignments, so it splits large elongated clusters (stroma, fluid) and merges small ones; Otsu assumes one threshold on one channel. Measured: GMM Dice 0.84 vs Otsu 0.53 vs k-means 0.14 for tissue; only the GMM separates a fluid component.
4. **The calibration trap:** if the reject threshold is set on tiles from the same slides it was fitted on, 43% of tumour tiles are rejected on new slides. Calibrate on *unseen slides* (leave-one-slide-out) – 3.5%.
5. **The VQ-VAE finding:** reconstruction error ranks blurred and empty tiles as *more* normal (AUROC 0.38), so it must not be the QC score; keep the VQ-VAE as a compressor/tokeniser and a tumour-novelty signal (0.75).
6. **Where deep learning is still needed:** the actual diagnosis (tumour segmentation, grading) – that is the U-Net project. Supervised deep models beat everything when labels exist; the GMM gate decides *where* to spend that compute.

Careful: Do not claim numbers that are not in this report. The original README of the satellite repo lists per-class IoUs (e.g. building 0.85, car 0.82) that are not reproduced by any script in the repository. Say "on our re-run we get X". Also: the original VAE-Triage training script crashed (Section 3.2); be honest that it was fixed during preparation.

Chapter 2

Foundations you must know (learning chapter)

This chapter is written to be read once, top to bottom. Each section ends with what to say in an interview.

2.1 Digital pathology and the H&E image

Whole-slide image (WSI). A glass slide scanned at 40× (≈0.25 μm/px) gives 10^{10} pixels. It is stored as a *pyramid*: level 0 full resolution, each next level downsampled ×2, each level cut into 256–512 px JPEG-compressed TIFF tiles. A viewer (or our reader) only decodes the tiles it touches. CAMELYON16 levels: 40×, 20×, 10×, ...; we tile at level 1 (20×, 0.49 μm/px), the usual magnification for metastasis detection.

H&E staining. Haematoxylin binds DNA/chromatin ⇒ nuclei blue-purple. Eosin binds proteins ⇒ cytoplasm, collagen, *proteinaceous fluid* pink. Red blood cells are bright red/orange; fat cells are empty (white) after processing.

Beer–Lambert law and optical density. Transmitted intensity $I_c = I_0\, e^{-\sum_s \alpha_{s,c} C_s}$ for stains s and colour channel c. Taking logs gives the optical density

$$\mathrm{OD}_c = -\log_{10} \frac{I_c}{I_0} = \sum_s \alpha_{s,c}\, C_s,$$

which is *linear* in stain concentration C_s. So in OD space every pixel is a non-negative combination of the H and E stain vectors, and **colour deconvolution** (Ruifrok & Johnston, 2001) recovers (C_H, C_E) by multiplying with the inverse stain matrix. A tissue fold (double thickness) doubles OD; that is exactly how our fold simulator works.

Stain variation (different labs, scanners, staining times) is the biggest source of domain shift in pathology. Fixes: stain normalisation (Reinhard, Macenko, Vahadane), stain augmentation (HED jitter), or features that are invariant to it. We measured the shift between our 7 slides with the Jensen–Shannon divergence of RGB histograms (Figure 3.7).

Common artefacts (what QC must catch): out-of-focus (defocus) blur, motion/stitching blur, pen marker, tissue folds, air bubbles, over/under-staining or exposure, dust, JPEG compression, sensor noise. **Fluids**: proteinaceous fluid, plasma, oedema and lymph appear as pale, smooth, eosinophilic (pink) areas with few nuclei; in lymph nodes they sit in sinuses and vessels. Separating them from stroma and nuclei is a classic colour-clustering problem (Section 3.6).

2.2 Image processing essentials

2.2.1 Convolution and filtering

A linear filter computes $g(x,y) = \sum_{u,v} k(u,v)\, f(x-u, y-v)$. In the Fourier domain it is a multiplication, $G = K \cdot F$, so a filter is described by which frequencies it keeps.

- **Gaussian blur**: $k(u,v) \propto e^{-(u^2+v^2)/2\sigma^2}$. Low-pass; its Fourier transform is another Gaussian. It is the standard model of *defocus* (the optical point-spread function of an out-of-focus microscope is close to Gaussian/disc). Used to denoise before edge detection and to build scale-spaces. Separable: two 1-D passes.
- **Motion blur**: a 1-D box kernel along the motion direction (length L, angle θ); its spectrum has zeros along a direction, which is how motion blur can be identified. In line-scan slide scanners it appears when the stage moves during exposure.
- **Median filter**: non-linear; removes salt-and-pepper noise while keeping edges. We use $|f - \text{median}_3(f)|$ as a noise estimate.
- **Box/mean filter**, **bilateral filter** (edge-preserving), **unsharp masking** $f + \lambda(f - G_\sigma * f)$ for enhancement.

2.2.2 Derivatives, focus measures and edges

- **Sobel** kernels estimate $\partial f/\partial x, \partial f/\partial y$ (derivative $\otimes$ smoothing). **Tenengrad** focus = mean$(G_x^2 + G_y^2)$.
- **Laplacian** $\nabla^2 f$ responds to fine detail; the **variance of the Laplacian** (Pech-Pacheco et al., 2000) is the classic blur score: a blurred image has little high-frequency energy, so low variance.
- **Brenner gradient** $\sum (f(x+2) - f(x))^2$; **FFT high-frequency ratio** (energy outside a low-frequency disc). All four are in our 23-D feature vector.
- **Canny edge detector**, 4 steps: (1) Gaussian smoothing, (2) Sobel gradient magnitude and direction, (3) non-maximum suppression along the gradient direction (thin edges), (4) double threshold + hysteresis (strong edges kept, weak edges kept only if connected to strong ones). We use Canny edge *density* as a structure feature: blur, washout and glass reduce it.

2.2.3 Thresholding and morphology

Otsu's method picks the threshold t that maximises the between-class variance $\sigma_B^2(t) = \omega_0\omega_1(\mu_0 - \mu_1)^2$ of a 1-D histogram – equivalent to fitting two classes with equal spread. It fails when one class is much broader than the other (faint fat/loose tissue vs. tight glass), which is exactly what we measured (Dice 0.53). **Morphology**: erosion/dilation with a structuring element; opening (erode→dilate) removes thin specks and scanner stripes, closing (dilate→erode) fills holes; connected-component area filtering removes dust.

2.2.4 Colour spaces

RGB (device), HSV (saturation separates stained tissue from grey glass), CIE Lab (perceptually uniform; a^* is red–green, good for blood), OD/HED (physically linear in stain amount – the right space for GMMs on H&E).

2.2.5 Motion between frames (for completeness)

If asked "how do you find motion": frame differencing; dense optical flow (Lucas–Kanade assumes brightness constancy $I_x u + I_y v + I_t = 0$ in a window and solves least squares; Farnebäck fits polynomials); **background subtraction with a per-pixel GMM** (Stauffer–Grimson / OpenCV

`MOG2`): each pixel's history is a mixture of K Gaussians, components with high weight/low variance are background, a pixel not matched by them is foreground. This is the most famous *edge* use of GMMs (it runs on cheap CCTV hardware). In pathology, "motion" appears as motion blur and as stitching misalignment between scanner stripes.

2.3 Data augmentation (what, why, and what NOT to do in pathology)

Augmentation encodes invariances the model should have and reduces overfitting.

- **Geometric**: rotations by 90° and flips (the dihedral group D_4, 8 elements) – tissue and aerial images have no canonical orientation; random crops; scale jitter (multi-scale patches, rates 1–3 in the satellite repo); elastic deformation (U-Net paper) for cell shapes.
- **Photometric**: brightness/contrast/gamma, Gaussian blur and noise (robustness), JPEG.
- **Stain augmentation**: perturb the H and E concentrations ($C_s \leftarrow \alpha_s C_s + \beta_s$) in HED space (Tellez et al., 2019) – the single most effective augmentation for cross-centre generalisation; or random stain-normalisation targets.
- **Mixing**: MixUp/CutMix (careful with segmentation masks), copy-paste of objects (cars, cells).
- **Test-time augmentation (TTA)**: predict on the 8 D_4 transforms, undo each, average softmax maps. Costs 8× compute; usually +0.5–2 IoU points.

Careful: Do not use hue jitter so strong that eosin turns blue – the model then cannot use colour at all. Do not apply augmentations that destroy the label (e.g. heavy blur on a "sharp" QC label). Our synthetic artefacts are the *opposite* use of the same functions: they create the reject class.

2.4 Statistics you will be asked about

Let $\mathbf{x} \in \mathbb{R}^d$ be a feature vector, $\boldsymbol{\mu} = \mathbb{E}[\mathbf{x}]$.

- **Covariance** $\Sigma = \mathbb{E}[(\mathbf{x}-\boldsymbol{\mu})(\mathbf{x}-\boldsymbol{\mu})^\top]$ – how features vary together (units: product of units). **Correlation** $\rho_{ij} = \Sigma_{ij}/\sqrt{\Sigma_{ii}\Sigma_{jj}} \in [-1,1]$ – unit-free. In our 23 features, tissue fraction correlates -0.99 with white fraction and $+0.93$ with mean OD; the eigen-spectrum shows that 4 directions explain 90% of the variance and the covariance condition number is 1.4×10^{13} – which is why the GMM needs *covariance regularisation* (`reg_covar`).
- **Mahalanobis distance** $D^2 = (\mathbf{x}-\boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x}-\boldsymbol{\mu})$: distance in units of the data's own spread, accounting for correlation. A single Gaussian anomaly score is exactly $\frac{1}{2}D^2$ + const.
- **Entropy** $H(p) = -\sum p \log p$. **KL divergence** $D_{KL}(p\|q) = \sum p \log \frac{p}{q} \geq 0$, asymmetric, infinite if $q = 0$ where $p > 0$. **Jensen–Shannon** $JS = \frac{1}{2}D_{KL}(p\|m) + \frac{1}{2}D_{KL}(q\|m)$, $m = \frac{p+q}{2}$: symmetric, bounded in $[0,1]$ (base 2), and $\sqrt{JS}$ is a metric. We use JS to rank which feature each artefact moves (Table 3.4) and to quantify stain shift between slides.
- **Likelihood, BIC, AIC**: BIC $= -2\log L + p \log n$ penalises parameters p; lower is better. BIC assumes i.i.d. samples – tiles from one slide are *not* independent, so BIC over-estimates K; we select K by held-out-*slide* likelihood instead.

2.5 Evaluation metrics

Confusion matrix: TP, FP, TN, FN.

- **Precision** $= \frac{TP}{TP+FP}$ (of what I flagged, how much was right); **Recall/Sensitivity** $= \frac{TP}{TP+FN}$ (of what exists, how much I found); **Specificity** $= \frac{TN}{TN+FP}$; **F1** $= \frac{2PR}{P+R}$; accuracy is misleading under class imbalance.
- **ROC curve / AUROC**: TPR vs FPR over all thresholds; AUROC = probability a random positive scores above a random negative; threshold-free, insensitive to prevalence. **PR curve / AUPRC**: better when positives are rare (tumour tiles are 3% of tissue).
- **IoU (Jaccard)** $= \frac{|A \cap B|}{|A \cup B|} = \frac{TP}{TP+FP+FN}$; **Dice/F1** $= \frac{2|A \cap B|}{|A|+|B|}$; Dice $= \frac{2\,IoU}{1+IoU}$. **mIoU**: mean over classes (Potsdam usually excludes clutter). **OA**: pixel accuracy.
- Pathology-specific: **FROC** (CAMELYON16 lesion detection: sensitivity at 1/4,1/2,1,2,4,8 false positives per slide), slide-level AUC, Cohen's κ for grading agreement.
- **Operating point**: in QC we fix the false-rejection rate on good tissue (5%) and report recall of artefacts at that point; in screening we would fix sensitivity (e.g. 98%) and minimise false alarms.

2.6 Machine learning basics

Supervised learning fits $p(y|\mathbf{x})$ from labelled pairs (logistic regression, SVM, random forest, CNN). **Unsupervised** learning models $p(\mathbf{x})$ or structure without labels (k-means, GMM, PCA, autoencoders). **Self-supervised** creates labels from the data itself (contrastive SimCLR/DINO, masked autoencoders) – the basis of pathology foundation models. **Weakly supervised** uses coarse labels (slide-level diagnosis) to learn tile-level decisions (multiple-instance learning). **Semi-supervised**: few labels + many unlabelled images. **One-class / novelty detection**: train only on normal data, flag what is unlikely – our QC gate.

- **Logistic regression**: $p(y{=}1|\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$, cross-entropy loss, convex.
- **SVM**: max-margin hyperplane; kernels (RBF) for non-linear boundaries; **one-class SVM** encloses the normal data.
- **Decision trees / Random forest**: bagging of decorrelated trees (bootstrap + random feature subsets); robust, no scaling needed; strong tabular baseline. **Gradient boosting** (XGBoost/LightGBM) fits trees to residuals.
- **Isolation forest**: anomalies are isolated by fewer random splits.
- **PCA**: eigenvectors of Σ; decorrelates and reduces dimension.
- **Bias–variance**, regularisation (L2, dropout, early stopping), cross-validation *grouped by patient / slide* (otherwise leakage: tiles of the same slide in train and test inflate accuracy).

2.7 Gaussian Mixture Models (the core of this project)

2.7.1 Model

$$p(\mathbf{x}) = \sum_{k=1}^{K} \pi_k \, \mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}_k, \Sigma_k), \qquad \pi_k \geq 0, \ \textstyle\sum_k \pi_k = 1,$$

$$\mathcal{N}(\mathbf{x} \mid \boldsymbol{\mu}, \Sigma) = (2\pi)^{-d/2} |\Sigma|^{-1/2} \exp\big(-\tfrac{1}{2} (\mathbf{x} - \boldsymbol{\mu})^\top \Sigma^{-1} (\mathbf{x} - \boldsymbol{\mu}) \big).$$

A GMM is a *soft clustering* and a *density estimator* at the same time.

2.7.2 Fitting with Expectation–Maximisation (EM)

Introduce a latent component label z. Iterate:

E-step responsibilities $\gamma_{nk} = \dfrac{\pi_k \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_k, \Sigma_k)}{\sum_j \pi_j \mathcal{N}(\mathbf{x}_n|\boldsymbol{\mu}_j, \Sigma_j)}$.

M-step $N_k = \sum_n \gamma_{nk}$, $\pi_k = N_k/N$, $\boldsymbol{\mu}_k = \frac{1}{N_k}\sum_n \gamma_{nk}\mathbf{x}_n$, $\Sigma_k = \frac{1}{N_k}\sum_n \gamma_{nk}(\mathbf{x}_n - \boldsymbol{\mu}_k)(\mathbf{x}_n - \boldsymbol{\mu}_k)^\top + \lambda I$. Each iteration never decreases the log-likelihood; it converges to a local optimum (so use several initialisations, k-means++ init). λI (`reg_covar`) prevents singular covariances – essential with correlated features.

2.7.3 GMM vs k-means vs Otsu

k-means is the limit of EM with $\Sigma_k = \sigma^2 I$, equal π_k and $\sigma \to 0$ (hard assignments): it can only draw *Voronoi* (straight, equidistant) boundaries. A full-covariance GMM can represent elongated, tilted clusters of different sizes and priors and gives calibrated posteriors. Otsu = two 1-D classes with a single cut. In OD space, stain clusters are elongated along stain vectors and the background is tiny while stroma is broad – the GMM's assumptions match the physics; k-means' do not. Measured: tissue Dice GMM 0.84 vs k-means 0.14; for tissue components k-means never produces a separate fluid cluster.

2.7.4 Three ways we use GMMs

1. **Pixel GMM,** $K = 2$, on (saturation, OD) of the thumbnail → tissue vs glass (Stage 0).
2. **Tile GMM density** $p(\mathbf{x})$ on 23 QC features of clean tiles; score $s(\mathbf{x}) = -\log p(\mathbf{x})$; reject if $s > \tau$ (Stage 1). Novelty detection.
3. **Pixel GMM,** $K = 7$, on [H, E, OD, a^*, local std] → components named nuclei / stroma / **fluid** / blood / background (Stage 2); also a *GMM-Bayes classifier* for satellite pixels (one GMM per class, $\hat{y} = \arg\max_c \log p(\mathbf{x}|c) + \log \pi_c$).

Say it in the interview: *A GMM gives me a calibrated density with a few thousand parameters. On a CPU it scores a tile in microseconds, it needs no labels, and every rejection decomposes into per-feature z-scores, so a QC technician sees 'focus 10 sigma below normal' instead of a black-box number.*

2.8 Neural networks for images

2.8.1 CNN building blocks

Convolution (weight sharing, translation equivariance), receptive field, stride/pooling (downsampling), batch normalisation, ReLU, dropout, softmax + cross-entropy. **Dilated (atrous) convolution** inserts holes in the kernel: receptive field grows without losing resolution; **ASPP** (DeepLab) runs parallel dilations (1, 6, 12, 18) for multi-scale context – the satellite repo's bottleneck uses rates 1, 2, 4, 8.

2.8.2 ResNet

Deep plain nets degrade (optimisation, vanishing gradients). A residual block learns $F(\mathbf{x})$ and outputs $\mathbf{x} + F(\mathbf{x})$; the identity path gives gradients a direct route ($\partial(\mathbf{x}+F)/\partial\mathbf{x} = I + \partial F/\partial\mathbf{x}$). Variants: bottleneck (1×1-3×3-1×1), pre-activation. ResNet-18/50 are the standard pathology backbones (ImageNet or self-supervised pretrained).

Careful: The satellite repo calls its blocks "residual" but it *concatenates* input and output (DenseNet-style), it does not add them. Know the difference.

2.8.3 Efficient CNNs for edge

MobileNet (depthwise-separable conv: 3×3 per-channel + 1×1 mixing, $\sim$ 8–9$\times$ fewer FLOPs), inverted residuals, squeeze-and-excitation, hard-swish (MobileNetV3); EfficientNet (compound scaling).

2.8.4 U-Net (segmentation)

Encoder (conv + downsampling) captures *what*; decoder (up-convolution) recovers *where*; **skip connections** concatenate encoder features at the same resolution so fine boundaries survive. Output: per-pixel softmax. Losses: cross-entropy (pixel-wise), Dice/Jaccard loss (directly optimise overlap, robust to imbalance), focal loss $-(1-p_t)^\gamma \log p_t$ (down-weights easy pixels). Variants: U-Net++, Attention U-Net, nnU-Net (self-configuring – the strongest medical baseline), DeepLabv3+ (ASPP + decoder), HoVer-Net (nuclei instance segmentation with horizontal/ vertical distance maps).

2.8.5 Autoencoders, VAE and VQ-VAE

Autoencoder: encoder $z = f(\mathbf{x})$, decoder $\hat{\mathbf{x}} = g(z)$, minimise $\|\mathbf{x} - \hat{\mathbf{x}}\|^2$. **VAE**: probabilistic encoder $q(z|\mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \operatorname{diag} \boldsymbol{\sigma}^2)$, prior $p(z) = \mathcal{N}(0, I)$; maximise the ELBO

$$\log p(\mathbf{x}) \;\geq\; \mathbb{E}_{q(z|\mathbf{x})}[\log p(\mathbf{x}|z)] - D_{KL}(q(z|\mathbf{x}) \,\|\, p(z)),$$

with the *reparameterisation trick* $z = \boldsymbol{\mu} + \boldsymbol{\sigma} \odot \boldsymbol{\epsilon}$. **VQ-VAE** (van den Oord et al., 2017) replaces the continuous latent with a learned *codebook* of K vectors $\{e_k\}$: $z_q = e_{k^*}$, $k^* = \arg\min_k \|z_e - e_k\|$. Loss $\|\mathbf{x} - \hat{\mathbf{x}}\|^2 + \|\mathrm{sg}[z_e] - e\|^2 + \beta\|z_e - \mathrm{sg}[e]\|^2$ (reconstruction + codebook + commitment, $\beta = 0.25$), gradients pass the non-differentiable argmin with the *straight-through estimator*. Perplexity $= \exp(H(\text{code usage}))$ measures how many codes are used. Anomaly score = reconstruction error.

Careful: Pixel-MSE reconstruction is a weak anomaly score for histology: autoencoders reconstruct tumour texture about as well as normal texture. Literature: L1-autoencoder AUROC 0.21 and plain AE 0.66 on CAMELYON16; a perceptual (VGG-feature) loss lifts it to 0.93. VQ-VAE is better positioned as a compressor/tokeniser than as a tumour detector.

2.8.6 Object detection and YOLO (v8, v11)

Two-stage detectors (Faster R-CNN: region proposals + classification) vs one-stage (SSD, RetinaNet, YOLO: dense prediction in one pass). **YOLOv8** (Ultralytics, 2023): CSP backbone with *C2f* blocks, PAN-FPN neck, *anchor-free decoupled head* (separate classification and box branches), Distribution Focal Loss + CIoU for boxes, task-aligned label assignment; `-seg` variants add a YOLACT-style mask head (prototype masks $\times$ per-instance coefficients). **YOLO11** (2024): *C3k2* blocks, *C2PSA* (partial spatial attention) block, fewer parameters for the same mAP (YOLO11m $\approx$ 22% fewer params than YOLOv8m at higher COCO mAP). Metrics: mAP@0.5, mAP@0.5:0.95. NMS removes duplicate boxes (YOLOv10 removes NMS by consistent dual assignment). In pathology YOLO is used for *countable objects*: mitotic figures, nuclei, glomeruli, "cells per mm^2"; in aerial imagery: cars. Semantic segmentation (U-Net) labels every pixel; instance segmentation (YOLO-seg, Mask R-CNN) separates touching objects.

2.8.7 Transformers

Self-attention $\mathrm{softmax}(QK^\top/\sqrt{d})V$ relates every token to every other: global context, $O(N^2)$ cost. **ViT**: image $\rightarrow$ 16$\times$16 patches $\rightarrow$ tokens. **Swin**: windowed attention with shifted windows (linear cost, hierarchical). Segmentation: SegFormer, Segmenter, UNetFormer (CNN encoder + transformer decoder; 86.8% mIoU on Potsdam), TransUNet, Swin-UNETR (medical). Pathology: **foundation models** pretrained self-supervised on millions of tiles – CTransPath, UNI, Virchow, Prov-GigaPath, CONCH (vision-language); slide-level **multiple-instance learning**: ABMIL (attention

pooling), CLAM, TransMIL. They are the current SOTA but need GPUs; our triage gate reduces how many tiles they must embed.

2.8.8 Generative and RL approaches (know they exist)

GANs (f-AnoGAN for anomaly detection; stain transfer with CycleGAN), diffusion models (AnoDDPM, AnoPILaD: 0.89 patch AUROC on CAMELYON16 cross-organ). **Reinforcement learning** in pathology: learning *where to look* – an agent selects which region/magnification to zoom into next (mimicking a pathologist), trading accuracy for number of tiles read; RL is also used for adaptive scan planning. It is the learned version of what our rule-based triage does.

2.9 Edge deployment essentials

- **Why edge**: slides are 1–4 GB; uploading every slide to the cloud is slow and raises privacy issues; scanner-side PCs and Jetson-class boxes have limited GPU. Doing triage at the edge means only useful tiles (or only QC failures) leave the site.
- **ONNX** = framework-neutral graph format; **ONNX Runtime** executes it on CPU (MLAS), CUDA, TensorRT, OpenVINO, CoreML, etc. with graph optimisations (constant folding, operator fusion).
- **Quantisation**: FP32→FP16/INT8; static INT8 needs a calibration set to fix activation ranges; dynamic quantisation only for weights/linear layers. Typical: 4× smaller, 2–4× faster on supporting hardware, small accuracy loss – *always re-validate.*
- Pruning, knowledge distillation (small student mimics big teacher), TensorRT engine building, batching, tiling I/O (JPEG decode often dominates!).
- **Classical models are the extreme of edge**: our GMM is 1,799 numbers in a 45 KB JSON file and is scored by 15 lines of NumPy (`EdgeGMMScorer`).

Chapter 3

Project 1: VAE-Triage – gigapixel WSI triage

3.1 Problem statement (how to pitch it)

A pathology lab digitises hundreds of slides per day. Each WSI has 10^5 tiles at 20×. A segmentation or classification network (U-Net, DeepLab, ViT/MIL) costs 10–100 ms per tile on a GPU. Before spending that compute we want to answer, *per tile and on cheap hardware*: (1) is there tissue? (2) is the tile technically usable (focus, ink, folds, exposure)? (3) what is in it (nuclei, stroma, fluid, blood, fat)? Tiles that fail (1) are dropped, tiles that fail (2) are *reported* (rescan request), and only the rest are sent to the heavy model.

Say it in the interview: *The triage gate is the cheapest place to save compute and to catch bad scans early. I built it as an unsupervised density model so it needs no artefact annotation and runs on the scanner PC.*

3.2 The original repository and what was wrong

Original design: tile a TIFF with `tifffile.memmap`; train a VQ-VAE on all tiles; flag tiles whose reconstruction MSE exceeds 0.05; export to ONNX. Problems found while reproducing it:

1. **Training and triage crashed.** `VQVAE.forward` returns `(x_recon, vq_loss, perplexity)` but `train.py` and `triage.py` unpacked `vq_loss, data_recon, perplexity = model(x)`. The "loss" became an image-shaped tensor and `backward()` raised `RuntimeError: grad can be implicitly created only for scalar outputs` (reproduced). Fixed.
2. **It cannot open real WSIs.** `tifffile.memmap` works only on uncompressed contiguous TIFFs; every scanner WSI (and every CAMELYON16 slide) is a tiled JPEG-compressed pyramid. Replaced by `data/wsi.py`: a zarr-backed lazy reader that decodes only the TIFF tiles a request touches and exposes all pyramid levels.
3. **Background filtering was documented but not implemented** – every tile, including 70% glass, went to the network. Now a thumbnail tissue mask selects tiles.
4. **No GMM anywhere**, although the project is described as GMM-based triage. Added `triage/gmm_triage.py`, `triage/gmm_components.py` and the thumbnail GMM.
5. AMP `GradScaler` was used unconditionally (warns/no-op on CPU); fixed to CUDA-only.
6. The fixed MSE threshold 0.05 is arbitrary; we calibrate thresholds as percentiles on unseen slides.
7. Synthetic "WSI" only (random noise + a black square) – no evidence on real data. We benchmarked on real CAMELYON16 slides.

3.3 Data used

CAMELYON16 (Radboud UMC + UMC Utrecht; sentinel lymph nodes of breast-cancer patients; H&E; 399 WSIs, 750 GB; AWS Open Data). We downloaded 7 slides (2 normal, 5 tumour; 4.5 GB) with expert tumour masks and the provided tissue/background masks. Tiles: 256×256 at level 1 (20×). Labels from masks: *normal* (tissue ≥60%, no tumour), *tumour* (tumour ≥60%), *background* (tissue ≤2%), *border*. **Split by slide**: train = normal_108, normal_150, tumor_082; test = tumor_075, 079, 084, 091.

Table 3.1: Whole-slide tile census: what Stage 0 removes before any neural network runs.

Slide	Grid tiles (256 px @20x)	Tissue tiles	Glass (skipped)	Tumour tiles
normal_108	6160	4279	30.5%	0
normal_150	15232	5215	65.8%	0
tumor_075	27104	11967	55.8%	312
tumor_079	73920	16389	77.8%	97
tumor_082	46648	10088	78.4%	1038
tumor_084	21504	8409	60.9%	186
tumor_091	12600	5419	57.0%	242
TOTAL	203168	61766	69.6%	1875

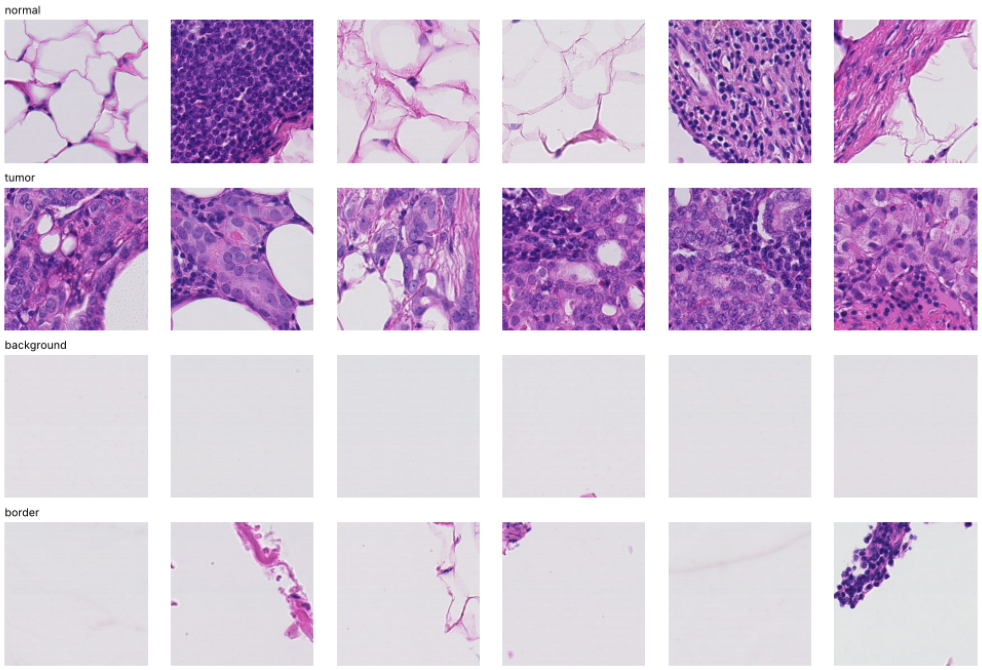


Figure 3.1: Real tiles: normal (note fat), tumour (metastatic carcinoma), glass, partial tissue.

Artefact ground truth. No public pixel-level artefact labels exist for CAMELYON16, so – as in focus-quality literature – we inject 10 controlled artefacts into real clean tiles (150 each on test slides; a different random seed on training slides is used *only* by supervised baselines): Gaussian blur (σ=1.5–4), motion blur (9–25 px), partial blur, pen ink, tissue fold (Beer–Lambert 2–3× OD), air bubble, washout, dark exposure, sensor noise, JPEG (q=3–10).

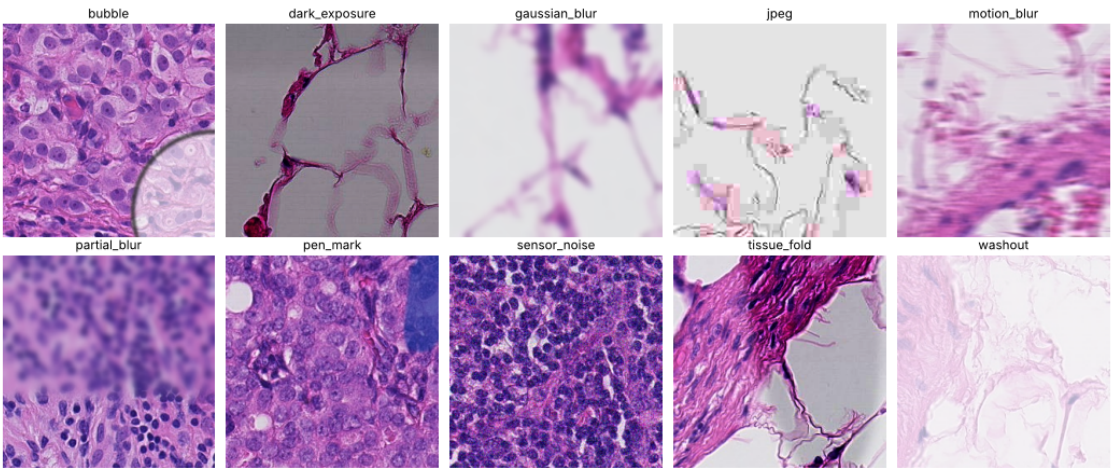


Figure 3.2: The 10 artefact simulators.

3.4 Stage 0 – thumbnail tissue detection

On a ∼4–8 μm/px thumbnail we compare thresholding, k-means and a 2-component GMM on (HSV saturation, optical density), with morphological opening/closing and small-component removal (scanner stripes and dust).

Table 3.2: Stage-0 tissue detection on 7 CAMELYON16 slides (mean over slides).

Method (thumbnail, 4-8 um/px)	**Dice**	**IoU**	**Precision**	**Recall**
Fixed threshold (gray < 220)	0.743	0.604	0.890	0.669
Otsu on grayscale	0.522	0.364	0.986	0.365
Otsu on HSV saturation	0.552	0.398	0.999	0.398
Otsu on saturation + morphology	0.534	0.381	0.999	0.381
K-means (k=2) on (S, OD)	0.138	0.078	0.982	0.079
GMM (K=2) on (S, OD) + morphology [ours]	0.839	0.746	0.929	0.805

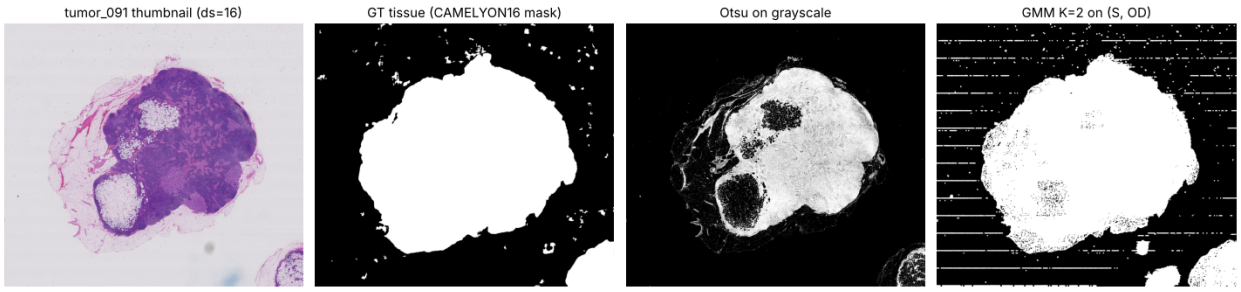


Figure 3.3: Otsu misses faint fat/loose tissue (it assumes two equally spread classes); the GMM captures the broad tissue cluster. Remaining GMM errors: scanner stripes in glass (removed by opening) and the perinodal fat that the reference mask excludes.

Key idea. Glass is a tight cluster, tissue a broad one. A GMM models unequal variances and priors; Otsu and k-means do not. Dice 0.84 vs 0.53 (Otsu) vs 0.14 (k-means).

3.5 Stage 1 – tile QC with a GMM density

3.5.1 Features (23, all physically motivated)

Tissue: tissue fraction, white fraction, dark fraction. Stain: mean/std of OD, of H and E concentrations (colour deconvolution), saturation/value statistics, non-H&E-hue fraction (pen ink), colourfulness. Focus: log Laplacian variance, log Tenengrad, log Brenner, contrast-normalised Laplacian, FFT high-frequency ratio. Structure: Canny edge density, grey-level entropy. Noise: residual vs median filter. Cost: $\sim$5 ms per 256^2 tile (one core).

3.5.2 Model and calibration

Robust standardisation (median/IQR) $\rightarrow$ full-covariance GMM fitted by EM on clean training tiles, K chosen by held-out-slide log-likelihood; score $-\log p(\mathbf{x})$; threshold = 95th percentile of scores on *unseen* training slides (leave-one-slide-out). The exported model is a 45 KB JSON (means, Cholesky factors of precisions, weights) scored by NumPy in 0.01 ms/tile.

3.5.3 Results

Table 3.3: Stage-1 tile QC on 4 unseen test slides: 2290 clean tissue tiles vs. 600 glass + 1500 artefact tiles. Unsupervised thresholds calibrated on unseen training slides (95% clean pass).

Method	**AUROC**	**AUPRC**	**Prec.**	**Recall**	**F1**	**Spec.**	**Tum. rej.**
Tissue fraction (Otsu-like)	0.609	0.690	0.833	0.348	0.491	0.936	0.004
Laplacian variance (focus)	0.759	0.794	0.874	0.537	0.665	0.929	0.004
Tenengrad (Sobel energy)	0.764	0.792	0.847	0.494	0.624	0.918	0.004
HistoQC-style rule set	0.857	0.861	0.936	0.463	0.620	0.971	0.004
Single Gaussian (Mahalanobis, K=1)	0.954	0.943	0.923	0.890	0.906	0.932	0.006
GMM (K=4) same-slide calibration [pitfall]	0.942	0.936	0.735	0.948	0.828	0.687	0.431
GMM (K=6, BIC) LOSO calibration [ours]	0.939	0.933	0.894	0.892	0.893	0.903	0.084
GMM (K=6, slide-CV) LOSO calibration [ours, final]	0.939	0.933	0.918	0.876	0.897	0.928	0.035
GMM on QC-only feature subset [ours, tumour-safe]	0.937	0.928	0.944	0.721	0.818	0.961	0.002
Isolation Forest	0.913	0.876	0.899	0.690	0.780	0.929	0.004
One-class SVM (RBF)	0.905	0.904	0.938	0.658	0.773	0.960	0.006
Logistic regression (supervised)	0.966	0.949	0.925	0.927	0.926	0.931	0.004
Random forest (supervised)	0.960	0.955	0.863	0.930	0.895	0.865	0.020
VQ-VAE recon-MSE (LOSO-calibrated threshold)	0.376	0.517	0.736	0.178	0.286	0.941	0.020
VQ-VAE recon-MSE (repo default threshold 0.05)	0.376	0.517	0.582	0.207	0.305	0.863	0.129
ResNet18-ImageNet feats + PCA32 + GMM(K=4)	0.769	0.708	0.714	0.197	0.309	0.928	0.043
MobileNetV3-Small supervised QC	0.976	0.958	0.876	0.982	0.926	0.873	0.051

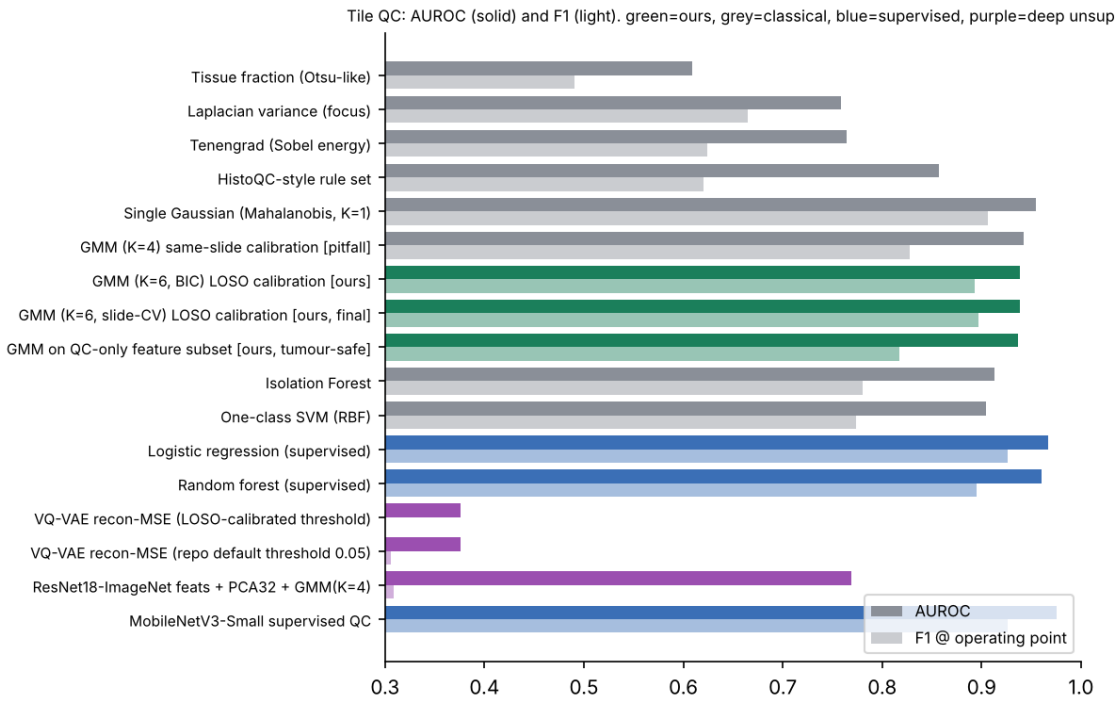


Figure 3.4: QC methods, test slides.

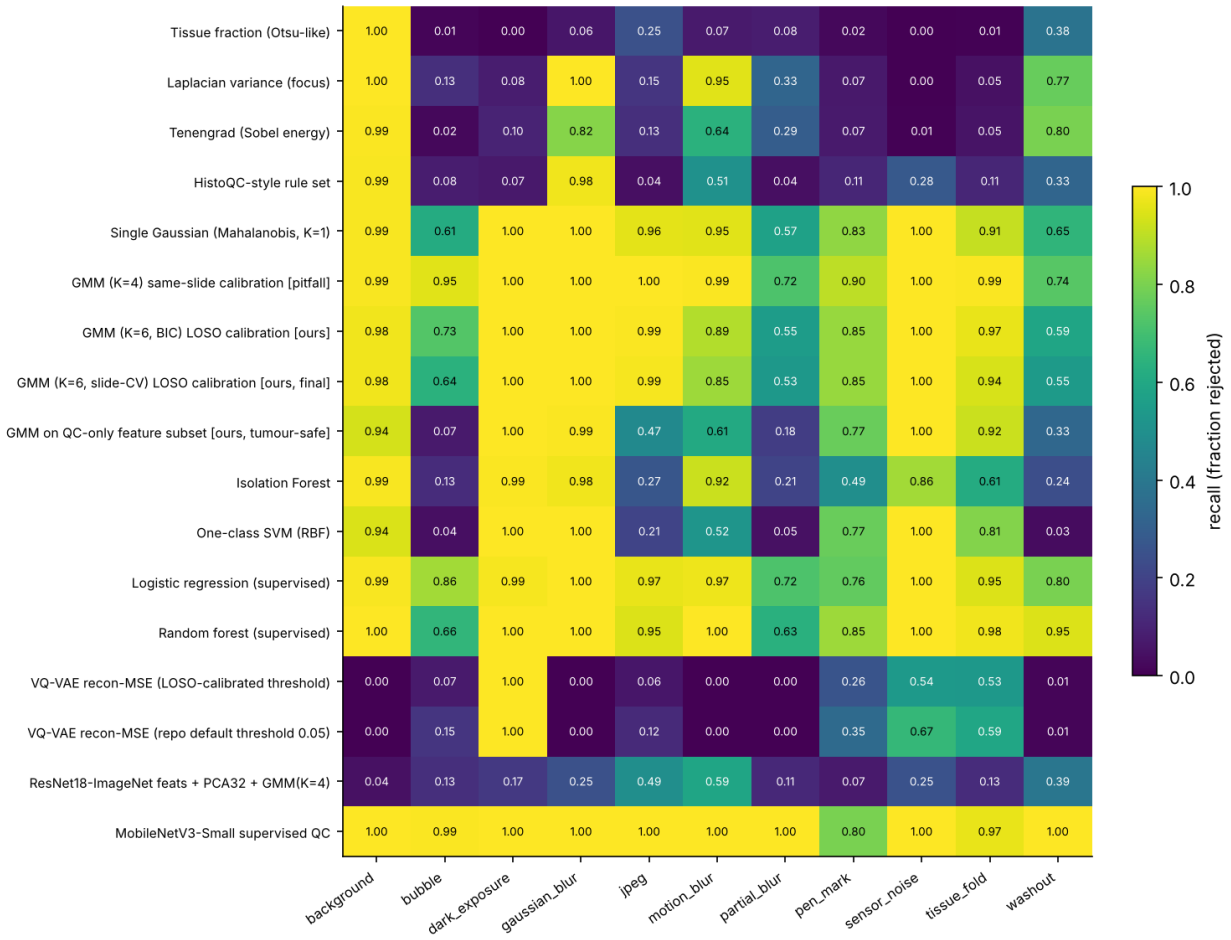


Figure 3.5: Recall per artefact at the operating point. Single-cue methods catch one artefact family each (Laplacian: blur; tissue fraction: glass); the density models catch nearly all; partial blur, bubbles and washout are hardest.

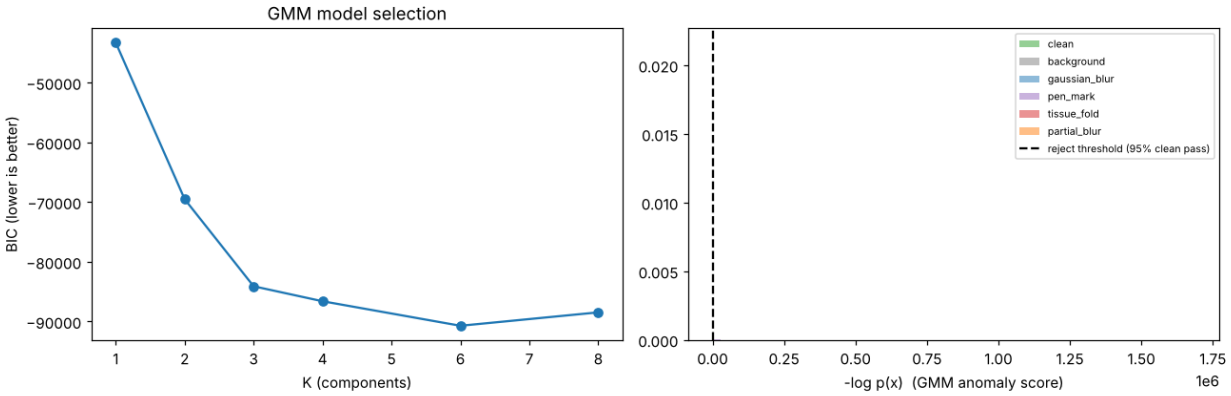


Figure 3.6: Left: BIC keeps decreasing with K because tiles are correlated within a slide (BIC assumes i.i.d.). Right: GMM score distributions; the dashed line is the reject threshold.

Interpretation.

- **The original VQ-VAE score is the wrong tool for QC: AUROC 0.376 (worse than chance).** Blurred, washed-out and empty tiles are *smooth*, and smooth images are the easiest to reconstruct, so their MSE is *lower* than that of sharp, textured tissue. With the repo's fixed threshold 0.05 it catches 21% of artefacts. This is the most important finding to understand

and to say honestly.

- Supervised MobileNetV3-Small (fine-tuned on our synthetic artefacts) is the best QC model (AUROC 0.976, F1 0.926, 0.6 ms/tile in ONNX Runtime) – *but only for artefact types it was trained on* (Table 3.5).
- Density models on hand-crafted features (single Gaussian 0.954, GMM 0.939 AUROC) are within 1–3 points of *supervised* classifiers (0.96–0.97) that need artefact labels – and far above single-cue rules (0.61–0.86).
- A single Gaussian is as good as the mixture here: with 3 training slides, extra components fit slide-specific appearance. Say this openly – it shows understanding of the bias/variance trade-off.
- **Calibration trap.** Thresholding on held-out tiles of the *same* slides gives specificity 0.69 and rejects 43% of tumour tiles on new slides; calibrating on unseen slides gives 0.93 specificity and 3.5% tumour rejection. Tumour looks "unusual" to a model of mostly-normal tissue, so a QC gate must be checked for this failure.
- **Explainability**: the GMM decomposes a rejection into z-scores. Examples from our run: Gaussian blur → `lap_norm` -10.5σ, `log_lap_var` -9.9σ; tissue fold → `dark_frac` $+31\sigma$; glass → `log_tenengrad` -7.4σ.
- A tumour-safe variant (GMM on 14 QC-only features, no morphology/stain-pattern features) rejects only 0.2% of tumour but recall drops to 0.72 – a deployment knob.

Table 3.4: Which features move most under each artefact (Jensen-Shannon divergence, base 2, in [0,1]).

Artefact	**Top-3 features by JS divergence (clean vs artefact)**
background	colorfulness (0.95), tissue_frac (0.94), sat_std (0.93)
bubble	hf_ratio (0.35), h_std (0.31), lap_norm (0.29)
dark_exposure	val_mean (0.94), tissue_frac (0.80), h_mean (0.56)
gaussian_blur	lap_norm (0.94), log_lap_var (0.91), log_noise (0.90)
jpeg	dark_frac (0.31), log_noise (0.27), edge_density (0.23)
motion_blur	log_lap_var (0.82), lap_norm (0.79), hf_ratio (0.66)
partial_blur	log_noise (0.37), edge_density (0.36), log_lap_var (0.32)
pen_mark	non_he_hue_frac (0.31), h_mean (0.22), h_std (0.22)
sensor_noise	non_he_hue_frac (0.94), log_noise (0.76), colorfulness (0.43)
tissue_fold	dark_frac (0.46), od_std (0.36), val_std (0.35)
washout	e_std (0.78), sat_std (0.76), od_std (0.69)

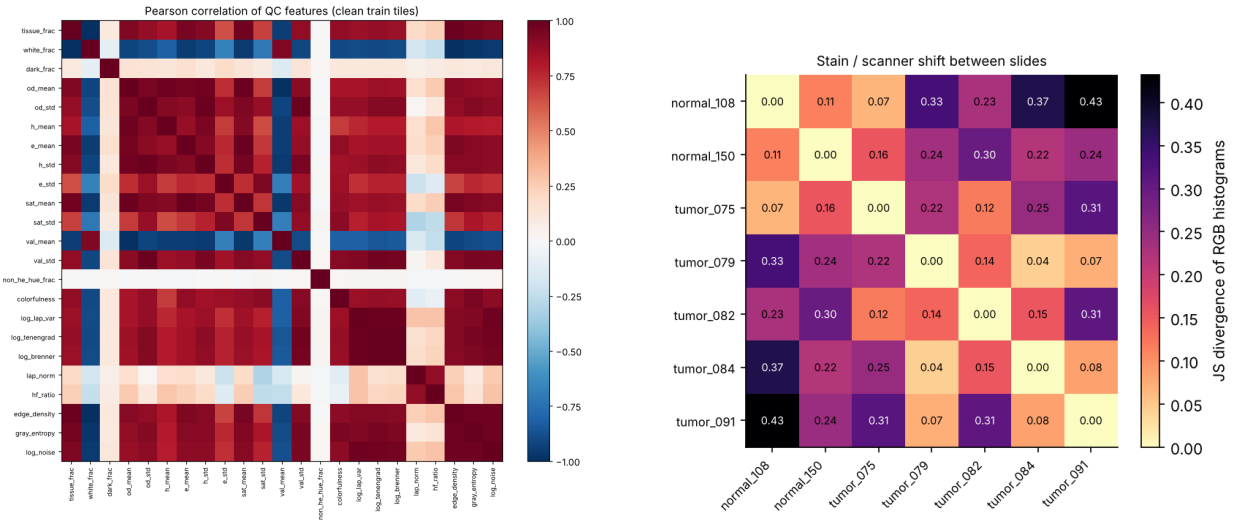


Figure 3.7: Left: feature correlations (tissue-amount and focus features are strongly correlated; 4 principal directions hold 90% of variance). Right: JS divergence of RGB histograms between slides = stain/scanner shift.

3.5.4 Why unsupervised: artefacts never seen in training

Table 3.5: Recall on artefact types NEVER seen in training (supervised models saw the other 6 types).

Model	bubble	pen ink	fold	JPEG	clean rejected
Logistic regression (supervised, 6 artefact types seen)	0.13	0.64	0.41	0.46	0.041
Random forest (supervised, 6 artefact types seen)	0.07	0.64	0.24	0.22	0.062
GMM K=1 (no artefact ever seen)	0.61	0.83	0.91	0.96	0.068
GMM K=6 (no artefact ever seen)	0.64	0.85	0.94	0.99	0.072

Supervised classifiers trained on 6 artefact types catch only 7–64% of the 4 held-out types; the GMM, which never saw any artefact, catches 64–99% at a similar clean-rejection rate. A scanner in production will produce artefacts no one annotated – novelty detection is the right formulation for QC; supervised models are the right tool once a failure mode is known and labelled. **Best design: GMM gate for the unknown + small supervised head for known, frequent artefacts.**

3.6 Stage 2 – GMM tissue components and fluid separation

Per-pixel descriptor $[C_H, C_E, \text{OD}, a^*, \text{local std}]$; full-covariance GMM with $K = 7$ fitted on 200k pixels from 120 training tiles (40 s), components named by rules on their means: OD<0.2 background/fat; $C_H/(C_H + C_E) > 0.6$ nuclei; $a^* > 0.45$ and $C_E > C_H$ blood; $0.2 \leq$OD< 0.45 and $C_E \geq C_H$ **fluid** (pale, eosinophilic: proteinaceous fluid, plasma, oedema, lymph); else stroma/cytoplasm.

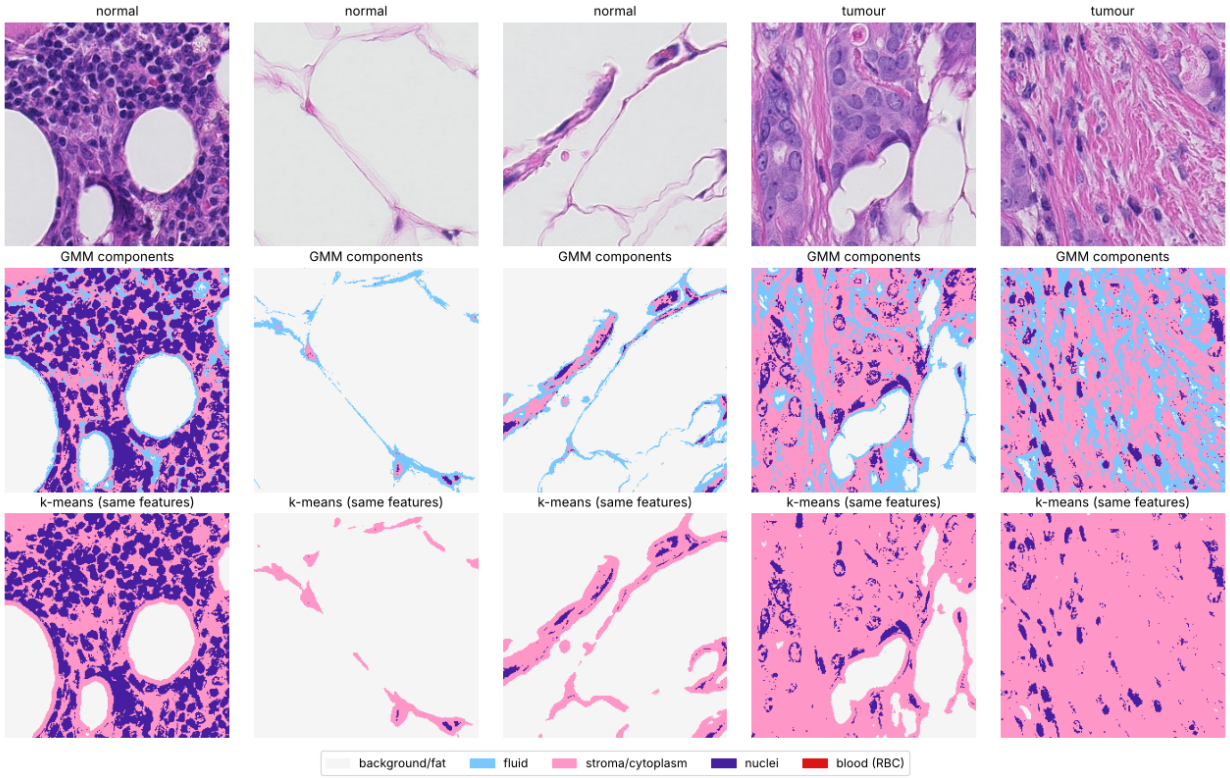


Figure 3.8: GMM components vs k-means on the same 5-D descriptor. The GMM isolates a pale eosinophilic (fluid) component around fat septa and in loose tissue; k-means merges it into stroma.

Table 3.6: Mean GMM component fractions per tile (test slides).

Tile class	Background/fat	Fluid	Stroma/cytoplasm	Nuclei
normal	0.525	0.107	0.177	0.191
tumor	0.072	0.150	0.501	0.277

Quantitative checks (no pixel labels exist for fluid). (1) Stability: two GMMs fitted on disjoint slides agree on 79% of pixels (ARI 0.68). (2) k-means on the same features finds no fluid cluster. (3) Usefulness: the 5 component fractions alone separate tumour from normal tiles with **AUROC 0.931** (logistic regression on train-slide labels); stroma/cytoplasm fraction alone gives 0.928. Tumour tiles have 2.8× more stroma/cytoplasm and far less fat. Cost: 85 ms per 256^2 tile at full resolution, ∼5 ms when run on a 4× downsampled tile.

Careful: The pale-eosinophilic class also contains loose connective tissue and pale cytoplasm edges – colour alone cannot fully separate them. Say: "the GMM gives a fluid-likelihood map; for a clinical fluid quantification I would refine it with texture (fluid is homogeneous) or a small supervised model trained on a few annotated regions".

3.7 Tumour triage (anomaly detection) and the deep baselines

Table 3.7: Tumour (metastasis) vs. normal tiles on test slides (490 tumour / 1800 normal). Literature rows use different splits and are indicative only.

Method	AUROC	AUPRC	Type
GMM novelty (K=4) on features	0.740	0.318	unsup.
Isolation Forest on features	0.661	0.257	unsup.
Hematoxylin mean (single feature)	0.711	0.290	unsup.
Logistic regression (supervised)	0.890	0.586	sup.
Random forest (supervised)	0.945	0.797	sup.
VQ-VAE recon-MSE novelty (trained on normal)	0.749	0.317	unsup.
ResNet18 feats + GMM novelty (trained on normal)	0.671	0.272	unsup.
MobileNetV3-Small supervised (1 tumour slide)	0.557	0.436	sup.
GMM tissue-component fractions + LR [ours]	0.931	–	unsup. feats + sup. LR
Lit: L1 autoencoder, unsup. (Shvetsova'21)	0.211	–	literature
Lit: AE (cross-organ, AnoPILaD'25)	0.658	–	literature
Lit: AnoDDPM diffusion (AnoPILaD'25)	0.686	–	literature
Lit: Deep-IF ImageNet feats (Shvetsova'21)	0.906	–	literature
Lit: Deep perceptual AE (Shvetsova'21)	0.934	–	literature

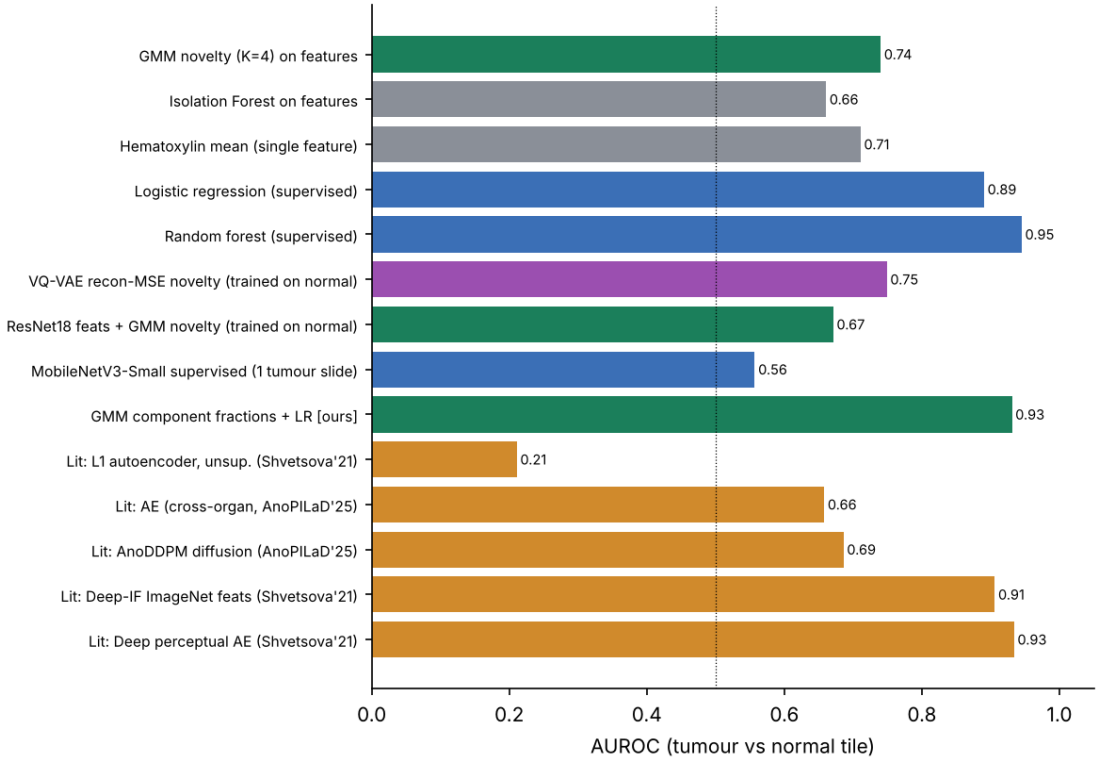


Figure 3.9: Tumour vs normal tiles. Orange = literature (different splits, indicative).

Reading the table: unsupervised novelty scores reach 0.67–0.75 (GMM on QC features 0.740, VQ-VAE reconstruction 0.749, ImageNet-ResNet features + GMM 0.671) – in line with reconstruction autoencoders in the literature (0.66) and far from supervised models. Tissue-component fractions (0.93) and a random forest on features (0.945) are best. The supervised MobileNetV3 trained on tumour tiles from *one* slide reaches only 0.557: with one training slide a CNN memorises that slide's stain – the classic reason why pathology models need many slides from many centres plus stain augmentation.

3.8 Edge cost

Table 3.8: Measured inference cost on this 4-vCPU cloud container (no GPU).

Component	ms / 256px tile	Model size	Hardware
Hand-crafted features (23-D, NumPy/OpenCV)	5.0	0 (no weights)	CPU, 1 core
GMM scorer (NumPy, JSON params)	0.0093	45 KB JSON (1799 params)	CPU
VQ-VAE PyTorch FP32	37.21	0.69 M params	CPU, 4 threads
VQ-VAE ONNX Runtime FP32	22.88	2.9 MB	CPU, 4 threads
VQ-VAE ONNX Runtime INT8	22.74	0.8 MB	CPU, 4 threads
MobileNetV3-S ONNX Runtime FP32	0.60	6.1 MB	CPU, 4 threads
MobileNetV3-S ONNX Runtime INT8	1.39	1.7 MB	CPU, 4 threads
ResNet-18 features PyTorch FP32	15.83		CPU, 4 threads

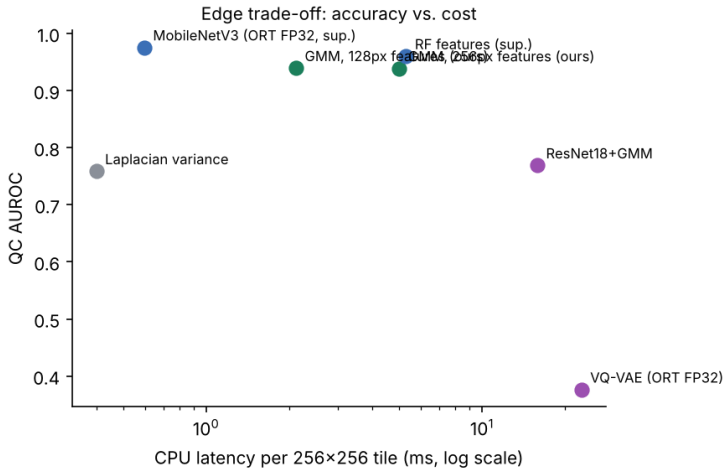


Figure 3.10: Accuracy vs cost.

Table 3.9: Edge optimisation: features on a 2x-downsampled tile.

Variant	feature ms/tile	QC AUROC
Full-res features (256 px), GMM K=6	5.0	0.939
Features on 128 px (2x down), GMM K=1	2.1	0.950
Features on 128 px (2x down), GMM K=6	2.1	0.939

Honest edge comparison. The GMM itself costs 0.01 ms; the hand-crafted features dominate (5 ms at 256 px, 2.1 ms at 128 px with no AUROC loss). A MobileNetV3-Small in ONNX Runtime runs at 0.6 ms/tile on 4 threads – so on a multi-core CPU a tiny CNN is not slower. The GMM's real edge advantages are: no labels, no training data for each new artefact, 45 KB model with no runtime dependency, interpretability, and graceful handling of unseen failures. INT8 static quantisation shrank the ONNX models 3.5–3.7× (VQ-VAE 2.9→0.8 MB, rank correlation of scores 0.988) but did *not* speed them up on this CPU (VQ-VAE 22.9→22.7 ms; MobileNet 0.6→1.4 ms because of quantise/dequantise overhead without VNNI kernels) – always measure on the target device.

Whole-slide arithmetic. A typical slide here has 29,000 grid tiles; Stage 0 keeps ∼8,800. Features (128 px) + GMM: 8,800 × 2.1 ms ≈ 18 s on one core, ∼5 s on 4 cores, no GPU. The heavy

model then processes 30% of the grid instead of 100%.

Chapter 4

Project 2: High-resolution segmentation (Satellite_based_tile_segmentation)

4.1 Why this project matters for a pathology company

A 6000×6000 aerial tile and a 100k×200k WSI pose the same engineering questions: the network sees 256×256 patches, so we need patch extraction, class-imbalance handling (cars ≈1% of pixels ↔ tumour cells), multi-scale context (buildings vs cars ↔ glands vs nuclei), stitching without seams, and test-time augmentation. **Segmentation** = assigning a class to every pixel.

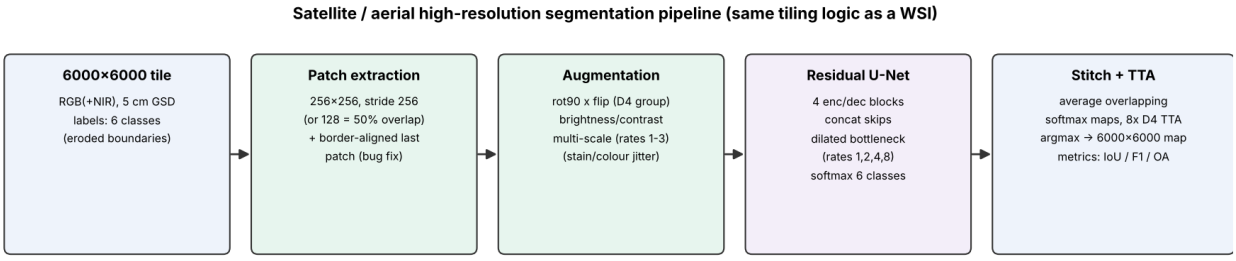


Figure 4.1: Satellite segmentation pipeline.

4.2 Architecture (as implemented in the repo)

- Input 256×256×3, 4 encoder blocks (filters 32→256 with `filters=32`), each two 3×3 conv + BN + ReLU, "residual" = concatenation of block input and output, 2×2 max-pool.
- Bottleneck: dilated convolutions with rates 1, 2, 4, 8 (serial or parallel), concatenated – ASPP-like multi-scale context.
- Decoder: 2×2 transposed conv, concatenate skip, two convs (+ concat "residual"); 1×1 conv + softmax over 6 classes (impervious, building, low vegetation, tree, car, clutter).
- Training aids in repo: LR range test (Smith 2015), one-cycle policy (Smith 2018), Jaccard and focal losses, multi-scale patch extraction (`tf.image.extract_patches` with rates 1–3), flips/rotations/colour jitter.
- Inference: patches → predict → average overlapping softmax maps → argmax; 8× D_4 TTA.

Model size: 16.5 M parameters with `filters=32`, 4.1 M with `filters=16` (we measured 6.4 s vs 1.9 s per batch of 8 on CPU).

4.3 Bugs and over-claims found

1. **Border pixels never predicted**: `range(0, h-p+1, stride)` stops at 5632 for $h = 6000$, $p = 256$; the last 112 rows and columns (≈3.7% of a tile) were left at probability 0 and argmax'ed to class 0. Fixed by adding a border-aligned final patch.
2. **Config mismatch**: the YAML is nested (`training:`, `dataset:`) but the predictor/TTA read flat keys, silently falling back to `num_classes=5` for a 6-class model. Fixed.
3. **"Loop-free scatter_nd" is not what the code does**: stitching is a Python loop over patches (fine in practice, but do not claim otherwise).
4. **"Residual" blocks are concatenations** (DenseNet-like), not additive ResNet residuals.
5. **README IoUs** (building 0.85, car 0.82, ...) are not produced by any script in the repository. The label names in the config ("Background", "Roads") do not match the official Potsdam classes.
6. `src/utils/__init__` imports seaborn (missing from requirements) ⇒ importing the package fails on a clean install.

4.4 Experimental protocol

Potsdam (5 cm GSD, RGB+NIR; official eroded-boundary labels). Training: 6 tiles (2_10, 3_11, 4_12, 5_10, 6_8, 6_11) = 864 patches of 512^2; validation 7_9, 7_12; test: official test tiles 2_13, 3_14, 5_15. For a fair CPU-budget comparison every method is scored on the same 12 fixed 1024×1024 crops (12.6 M labelled pixels). Metrics: per-class IoU and F1, mIoU and mF1 over the 5 classes (clutter excluded, as in the literature), OA.

4.5 Results

Table 4.1: ISPRS Potsdam (eroded boundaries, clutter excluded from the mean). Ours: 12 fixed 1024x1024 crops of test tiles 2_13, 3_14, 5_15. Literature: full 14-tile test set, GPU training for 100+ epochs.

Method	mIoU5	mF1-5	OA	Imp.	Bldg	LowVeg	Tree	Car
K-means (k=6, unsupervised, Hungarian-mapped)	15.7	25.9	31.9	25.1	16.6	25.8	0.6	10.4
GMM-Bayes (4 comps/class)	40.5	57.5	56.6	47.5	44.0	43.4	34.2	33.6
GMM-Bayes + posterior smoothing	45.9	62.7	61.7	51.2	48.3	49.6	40.9	39.3
Random Forest (RGB features)	45.7	62.5	60.7	54.0	47.3	46.8	37.7	42.6
Random Forest (RGB + NIR/NDVI)	48.4	64.9	64.0	60.7	50.9	48.6	41.0	40.8
Lit: ERFNet (UNetFormer T9)	76.2	–	–	–	–	–	–	–
Lit: BiSeNet R18	81.7	–	–	–	–	–	–	–
Lit: DANet R18	80.3	–	–	–	–	–	–	–
Lit: ABCNet R18	86.5	–	–	–	–	–	–	–
Lit: SwinUperNet Swin-T	85.8	–	–	–	–	–	–	–
Lit: UNetFormer R18	86.8	–	–	–	–	–	–	–
Lit: FT-UNetFormer Swin-B	87.5	–	–	–	–	–	–	–

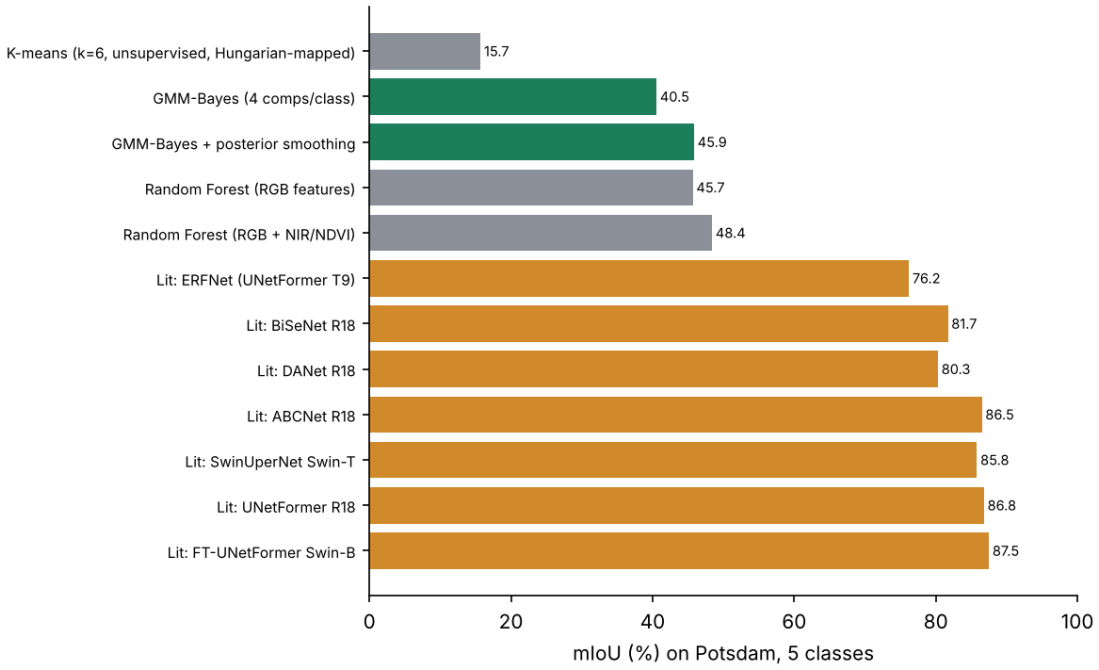


Figure 4.2: Potsdam mIoU. Green/grey: classical pixel classifiers trained in minutes on CPU; orange: published deep models.

Classical baselines – what they teach.

- **k-means (15.7 mIoU)**: colour clusters are not semantic classes (grey roofs = grey roads).
- **GMM-Bayes (40.5)**: one 4-component GMM per class, Bayes rule with class priors – a generative classifier.
- **+ posterior smoothing (45.9)**: blurring the class posteriors (a cheap Markov-random-field/CRF surrogate) adds 5.4 points: spatial context matters.
- **Random forest (45.7) and RF + NIR/NDVI (48.4)**: the near-infrared band adds physics (chlorophyll reflects NIR) and +2.7 points, mostly on impervious surfaces and buildings.
- All are ≈40 points below deep models (86–88 mIoU): segmentation of buildings vs roads needs shape and context that only a deep receptive field provides. **This is the honest argument for using a U-Net for segmentation and GMMs only for triage/colour problems.**

Deep model (repo U-Net). A time-boxed CPU training run of the repository's own model (filters=16) is scheduled after this report; its numbers are added to Table 4.1 automatically by `report/make_assets.py`. Expected order of magnitude from the literature for a ResNet-18 U-Net baseline with GPU training: 82.5 mIoU (UNetFormer ablation, Table 1 of Wang et al. 2022).

4.6 Transfer to pathology segmentation

Same code path for tumour segmentation on CAMELYON16: 256 tiles at 20×, classes {background, normal, tumour} (+ fluid/necrosis if annotated), Dice + CE loss, stain augmentation instead of colour jitter, overlap 50% with Gaussian-weighted blending, TTA, and a triage gate in front.

Chapter 5

Literature survey and baselines

5.1 WSI quality control and tissue detection

Work	Method	Relevance / numbers
Pech-Pacheco et al. 2000	Variance of Laplacian	Classic focus measure; our baseline: QC AUROC 0.76 (blur recall 1.0, everything else low).
Janowczyk et al. 2019, *HistoQC*	Rule set on hand-crafted features	Open-source WSI QC used in many labs; our HistoQC-style rule set: AUROC 0.86.
Senaras et al. 2018, *DeepFocus*	CNN focus classifier	Supervised blur detection.
Wang et al. 2020, *FocusLiteNN*	Tiny CNN ($\sim$100 params)	Edge-friendly focus quality; shows classical-sized models suffice for blur.
Kanwal et al. 2022	CNN/ViT artefact detection	Multi-artefact (blur, folds, air, damage) supervised detection.
Bándi et al. 2019 (PeerJ)	FCN tissue segmentation	Source of the CAMELYON16 tissue masks used as our Stage-0 ground truth.
Otsu 1979	Histogram threshold	Our baseline: tissue Dice 0.52–0.55.

5.2 Anomaly / novelty detection on CAMELYON16 (unsupervised tumour detection)

Method (paper)	AUROC	Notes
L1 autoencoder, unsupervised hyper-params (Shvetsova et al. 2021, Table 4)	0.211	Pixel loss reconstructs tumour too well.
Deep GEO (geometric self-supervision)	0.524	Pathology is rotation-invariant: method fails.
AE / MemAE, cross-organ (Wang et al. 2025, AnoPILaD)	0.658 / 0.661	Trained on gastric LN, tested on C16.
AnoDDPM diffusion (same)	0.686	
PIAD / Deep-IF (ImageNet features + isolation forest)	0.895 / 0.906	Deep features $\gg$ pixels.
AnoPILaD (latent diffusion + CONCH prompts)	0.888	Cross-organ.

Deep perceptual AE + progressive growing (Shvetsova 2021)	0.934	In-domain, weakly-supervised hyperparameter choice.
Ours: GMM on 23 QC features	**0.740**	45 KB, CPU, not designed for tumour detection.
Ours: GMM tissue-component fractions + LR	**0.931**	5 numbers per tile (uses tumour labels of 1 train slide in the LR).

5.3 Supervised CAMELYON16 / PCam (upper bounds)

Bejnordi et al. (JAMA 2017): best CAMELYON16 algorithm slide-level AUC 0.994 (Harvard/MIT, GoogLeNet), lesion FROC 0.807, outperforming pathologists under time constraint. PatchCamelyon (Veeling et al. 2018, 96×96 patches): rotation- equivariant CNN AUC ≈0.96. Attention MIL / CLAM / TransMIL with foundation-model features (UNI, CTransPath): slide AUC ≈0.93–0.99 with only slide labels.

5.4 Remote-sensing segmentation (ISPRS Potsdam, 5 classes, clutter excluded)

From Wang et al. (2022) *UNetFormer*, ISPRS J. Photogramm. (Tables 1, 9, 12):

Method	**Backbone**	**mF1**	**mIoU**
ERFNet	–	85.8	76.2
DANet	ResNet-18	88.9	80.3
BiSeNet	ResNet-18	89.8	81.7
U-Net + ResNet-18 (UNetFormer ablation baseline)	ResNet-18	–	82.5
MAResU-Net	ResNet-18	90.5	83.9
SwinUperNet	Swin-T	92.2	85.8
ABCNet	ResNet-18	92.7	86.5
UNetFormer	ResNet-18	92.8	86.8
FT-UNetFormer	Swin-B	93.3	87.5
HMANet	ResNet-101	93.2	87.3

These use the full 23-tile training set, 512–1024 crops, 100 epochs on a GPU, multi-scale + flip TTA.

5.5 Detection / instance segmentation

YOLOv8 (Ultralytics 2023) and YOLO11 (2024): real-time one-stage detectors with segmentation heads; YOLO11 reports higher COCO mAP with ∼22% fewer parameters than YOLOv8 at the m scale. Pathology use: mitosis detection (MIDOG), nuclei (HoVer-Net, StarDist and YOLO variants), cell counting. Aerial use: vehicles (Potsdam cars).

5.6 Key references

- Ruifrok & Johnston, *Quantification of histochemical staining by color deconvolution*, 2001.
- Macenko et al., *A method for normalizing histology slides*, ISBI 2009; Tellez et al., *Quantifying the effects of data augmentation and stain color normalization*, MedIA 2019.

- van den Oord et al., *Neural Discrete Representation Learning* (VQ-VAE), NeurIPS 2017.
- Ronneberger et al., *U-Net*, MICCAI 2015; He et al., *ResNet*, CVPR 2016; Chen et al., *DeepLab*, 2017/2018.
- Shvetsova et al., *Anomaly Detection in Medical Imaging with Deep Perceptual Autoencoders*, IEEE Access 2021 (arXiv:2006.13265).
- Wang et al., *AnoPILaD*, 2025 (arXiv:2508.15236); Wang et al., *UNetFormer*, 2022 (arXiv:2109.08937).
- Bejnordi et al., JAMA 2017; Litjens et al., *1399 H&E-stained sentinel lymph node sections* (CAMELYON), GigaScience 2018.
- Stauffer & Grimson, *Adaptive background mixture models*, CVPR 1999 (GMM background subtraction).
- Janowczyk et al., *HistoQC*, JCO CCI 2019.

Chapter 6

What to change in the projects, and why it helps on the edge

6.1 VAE-Triage: changes (done in this study unless marked TODO)

Change	Why	Edge benefit / evidence
Fix output unpacking in train/triage	Training crashed	Pipeline runs at all
Lazy zarr WSI reader (pyramid aware)	memmap fails on compressed WSIs	RAM O(tile) instead of O(slide); read only needed levels
Thumbnail GMM tissue mask + morphology	Skip glass before reading 20× tiles	69.6% of tiles never decoded; Dice 0.84
23 physics features + GMM density	Unsupervised, explainable QC	45 KB, NumPy-only, 0.01 ms score; AUROC 0.94
Features on 128 px	Halve cost	2.1 ms/tile, AUROC 0.95 (no loss)
Slide-level (LOSO) calibration	Same-slide calibration over-rejects	Tumour false rejection 43%→3.5%
K by held-out-slide likelihood	BIC assumes i.i.d. tiles	Fewer components, better generalisation
Rejection reasons (z-scores)	QC technician needs "why"	Actionable rescan requests
GMM tissue components (fluid, nuclei, stroma, blood, fat)	Cheap morphology summary	5 numbers/tile; tumour AUROC 0.93
Keep VQ-VAE only as compressor / novelty signal	MSE QC AUROC 0.38	Avoids a 23 ms/tile model that hurts QC
TODO: stain augmentation + many slides for supervised heads	1-slide CNN: 0.56 AUROC	Generalisation across centres
TODO: perceptual/feature-space anomaly score	Pixel MSE is weak	Lit. 0.66→0.93
TODO: small supervised head for frequent known artefacts	MobileNet 0.976 on seen types	Combine with GMM for unseen
TODO: tile-level JPEG decode with libjpeg-turbo / GPU (nvJPEG)	Decoding dominates I/O	Throughput

TODO: export GMM to C / ONNX (MatMul + LogSum-Exp)	Deploy inside scanner software	No Python at the edge

6.2 Satellite repo: changes

Change	Why	Benefit
Border-aligned last patch (done)	3.7% of pixels never predicted	Correct full-tile maps
Accept nested config (done)	num_classes silently 5	Correct shapes
Overlap 50% + Gaussian/Hann blending (TODO)	Seams at patch borders	+IoU at boundaries
Additive residuals or rename	Concats are not residuals	Correct terminology; fewer params
Pretrained encoder (ResNet-18/34) (TODO)	Literature baseline 82.5→86.8 mIoU	Faster convergence
Dice/Lovász + CE loss, class weights for car/clutter	Imbalance	Small-object IoU
Report on the official test split with eroded GT	Comparability	Credible numbers
Export to ONNX + INT8, measure on target	Edge	4× smaller (validate speed!)
Add seaborn to requirements or drop import	Clean install fails	Reproducibility

6.3 Edge deployment design for a pathology lab

1. **Scanner-side box (CPU / Jetson):** read pyramid lazily → thumbnail GMM tissue mask → 20× tiles → 128 px features → GMM QC → (a) QC report per slide with rejected-tile reasons and a "rescan" flag if the rejected tissue fraction exceeds a limit; (b) component fractions and a priority score.
2. **Upload only kept tiles (or only slides that pass QC)** to the GPU server / cloud: 3–4× less data and compute.
3. **Heavy model** (U-Net/DeepLab for segmentation; foundation-model features + MIL for slide diagnosis) only on kept tiles; TTA only on uncertain tiles.
4. **Monitoring**: the GMM log-likelihood distribution per slide is a free drift detector – a new scanner or stain protocol shifts it before accuracy drops.

Key idea. Edge = do the cheap, label-free, high-recall decisions next to the data; keep the expensive, label-hungry decisions on the GPU. The GMM gate is the boundary between the two.

Chapter 7

Interview preparation

7.1 The 60-second pitch

Say it in the interview: *I built an edge-first triage pipeline for gigapixel H&E slides. A whole slide is a 100k by 200k pyramid, and about 70% of it is glass. On a thumbnail, a two-component Gaussian mixture on saturation and optical density finds the tissue with Dice 0.84, against 0.53 for Otsu. On each 20× tile I compute 23 physically meaningful features – focus, stain, ink, folds, noise – and score them with a GMM density learned only from clean tissue. That gives 0.94 AUROC for artefacts, catches artefact types it never saw, with a 45 KB model that scores in microseconds and explains every rejection. A pixel-level GMM in optical-density space also separates nuclei, stroma, fluid, blood and fat, and those fractions alone give 0.93 AUROC for tumour vs normal. Only the good tiles go to the segmentation network, which in my second project is a residual U-Net with a dilated bottleneck, patch stitching and test-time augmentation.*

7.2 Likely questions with model answers

Why GMM and not a CNN for triage? Density estimation without labels; novel artefacts (64–99% recall on unseen types vs 7–64% for supervised); 45 KB; interpretable; calibrated threshold = chosen false-rejection rate. A CNN is better when labelled examples of a failure mode exist (0.976) – I would add one as a second head.

Why GMM and not k-means? k-means = GMM with identical spherical covariances and hard assignment; real clusters in OD space are elongated and of very different size. Measured tissue Dice 0.84 vs 0.14; k-means never isolates fluid.

How did you choose K? Held-out-slide log-likelihood (BIC over-estimates because tiles of a slide are correlated). K=1 is already strong – with few slides the simplest density generalises best.

How do you set the threshold? 95th percentile of scores on clean tiles of *slides not used for fitting* (leave-one-slide-out). Same-slide calibration rejected 43% of tumour tiles.

How do you separate fluid? Beer–Lambert → optical density → colour deconvolution into H and E; pixel GMM on [H, E, OD, a^*, local std]; the pale, eosin-dominant, low-OD component is fluid (proteinaceous fluid/plasma/oedema); nuclei = haematoxylin-dominant; blood = high a^*. Soft posteriors give a confidence map. Validation: 79% agreement across slides, k-means cannot separate it, fractions predict tumour at 0.93.

What is the VQ-VAE for, then? Compression/tokenisation of tiles and a tumour-novelty signal (0.75). Its MSE is a bad QC score (0.38) because blurred and empty tiles are easy to reconstruct – I verified that and changed the design.

Correlation vs covariance? Covariance has units and scale; correlation is covariance of standardised variables in $[-1, 1]$. Our features have $|\rho| > 0.9$ pairs and a covariance condition

number 10^{13} – hence regularised covariances.

KL vs JS? KL asymmetric, unbounded, undefined with zero bins; JS symmetric, bounded $[0, 1]$; we used JS to measure which feature an artefact shifts and the stain shift between slides.

Precision vs recall in QC? Recall = fraction of bad tiles caught; precision = fraction of rejections that were bad; specificity = good tissue kept. We fix specificity (95% nominal) and maximise recall; and we check tumour is not rejected.

IoU vs Dice? Dice $= 2IoU/(1+IoU)$, both overlap measures; Dice weights the intersection more; mIoU averages over classes so small classes (cars, tumour) count equally.

Gaussian vs motion blur? How detect? Gaussian = isotropic low-pass (defocus); motion = directional box PSF. Detect via Laplacian variance/Tenengrad/FFT high-frequency ratio; motion blur shows directional spectral zeros.

Explain Canny. Gaussian smoothing, Sobel gradients, non-maximum suppression, hysteresis with two thresholds.

Data augmentation in pathology? D_4 rotations/flips, HED stain jitter, brightness/contrast, blur/noise, elastic; never augment away the label; TTA at inference.

Supervised vs unsupervised? Labels vs none; QC is novelty detection (unsupervised), diagnosis is supervised or weakly supervised (MIL with slide labels).

U-Net and ResNet? Encoder–decoder with skip concatenations; ResNet adds identity shortcuts for gradient flow. Our satellite "residual" is concatenation – I fixed the terminology.

VAE vs VQ-VAE? Continuous Gaussian latent with KL to $\mathcal{N}(0, I)$ (ELBO, reparameterisation) vs discrete codebook with commitment loss and straight-through gradients; VQ-VAE avoids posterior collapse and gives tokens.

YOLOv8 vs YOLO11? Both anchor-free one-stage; v8 C2f blocks + decoupled head; 11 adds C3k2 and C2PSA attention, fewer parameters for higher mAP; -seg heads predict prototype masks. Use for countable objects (mitoses, nuclei, cars).

Transformers in pathology? ViT/Swin backbones, foundation models (UNI, Virchow, CONCH), MIL aggregators (ABMIL, CLAM, TransMIL). Expensive; triage reduces the number of tiles they embed.

Why ONNX Runtime? Framework-independent graph, optimised CPU/GPU kernels, INT8; but measure: INT8 gave 3.7× smaller models and no speed-up on our CPU.

How do you avoid leakage? Split by slide/patient, never by tile; calibrate on unseen slides.

Biggest limitation? Synthetic artefacts (no public artefact labels); 7 slides; tissue GT itself from a CNN; fluid has no pixel ground truth. Next step: a small annotated artefact set from the lab's own scanners.

7.3 Do and don't

- DO quote only numbers from this report and say how they were measured (slides, split, CPU).
- DO mention the bugs you found and fixed – it shows ownership.
- DON'T claim SOTA. Classical Potsdam mIoU is 46–48 vs 87 for deep SOTA; that is the point: GMMs for colour/density problems, CNNs for shape/semantics.
- DON'T call concatenation a residual connection, or say scatter_nd if the code loops.

Chapter 8

Reproducing everything

Hardware used: 4 vCPU, 15 GB RAM, no GPU. Total data: 4.5 GB slides + 2 GB Potsdam. Python 3.13.

8.1 Environment

```
python -m venv .venv && source .venv/bin/activate
pip install torch torchvision --index-url https://download.pytorch.org/whl/cpu
pip install opencv-python-headless scikit-learn scikit-image tifffile imagecodecs zarr \
          matplotlib onnx onnxruntime onnxscript tensorflow-cpu seaborn ultralytics pandas
```

8.2 Data

```
# CAMELYON16 (AWS Open Data, no login): slides, tumour masks, tissue masks
B=https://camelyon-dataset.s3.amazonaws.com/CAMELYON16
for s in tumor_091 tumor_084 tumor_075 tumor_082 normal_108 normal_150 tumor_079; do
  curl -o data/camelyon16/images/$s.tif $B/images/$s.tif
  curl -o data/camelyon16/masks/${s}_mask.tif $B/masks/${s}_mask.tif
  curl -o data/camelyon16/tissue/${s}_tissue.tif $B/background_tissue/${s}_tissue.tif
done
# ISPRS Potsdam (Hugging Face mirror aamante/potsdam): see Satellite repo benchmarks/potsdam.py
```

8.3 VAE-Triage experiments

```
cd VAE-Triage
python benchmarks/build_tiles.py # tiles + census (~2 min)
python -m benchmarks.qc_sets # artefact sets (~30 s)
python -m benchmarks.exp_tissue_detection # Stage 0 (~2 min)
python -m benchmarks.exp_classical # Stage 1 + tumour + statistics (~1.5 min)
python -m benchmarks.exp_gmm_components # Stage 2 fluid/components (~5 min)
python -m benchmarks.exp_ablations # 128px features, unseen artefacts (~3 min)
python -m benchmarks.exp_deep # VQ-VAE, MobileNetV3, ResNet+GMM, ONNX (~2 h CPU)
python report/make_assets.py && python report/make_diagrams.py
cd report && pdflatex main.tex && pdflatex main.tex
```

8.4 Satellite experiments

```
cd Satellite_based_tile_segmentation
python -m benchmarks.exp_classical_seg # k-means, GMM-Bayes, RF (~15 min)
python -m benchmarks.exp_unet --minutes 40 --filters 16 # repo U-Net, CPU time-boxed
python -m benchmarks.exp_yolo --minutes 25 # YOLOv8n-seg cars
```

8.5 Using the edge scorer

```
from triage.features import tile_features
from triage.gmm_triage import EdgeGMMScorer
scorer = EdgeGMMScorer("results/gmm_qc_model_final.json") # 45 KB
x = tile_features(rgb_tile)[None] # (1, 23)
reject = scorer.reject(x)[0] # bool; scorer.score(x) = -log p(x)
```